

# Mathematical Foundations

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These notes develop the core mathematical toolkit for the course: the language of probability, three generating functions (MGF, CGF, characteristic function), Gaussian integration in arbitrary dimension, Wick's theorem, the saddle-point approximation, and the Hubbard–Stratonovich transformation. Every definition is followed by a worked example; every theorem is derived step by step.

## Probability: basic definitions

A **sample space**  $\Omega$  is the set of all possible outcomes of an experiment. A **random variable**  $X$  is a function  $X : \Omega \rightarrow \mathbb{R}$  assigning a real number to each outcome. When  $X$  is continuous its distribution is described by a **probability density function** (pdf)  $f_X(x)$  satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1. \quad (1)$$

The probability that  $X$  falls in  $[a, b]$  is  $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$ . The **cumulative distribution function** (CDF) is  $F_X(x) = \int_{-\infty}^x f_X(t) dt$ , so  $f_X = F'_X$ . When  $X$  is discrete we write  $P(X = x_k) = p_k$  with  $\sum_k p_k = 1$ .

The **expectation** and **variance** are

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (2)$$

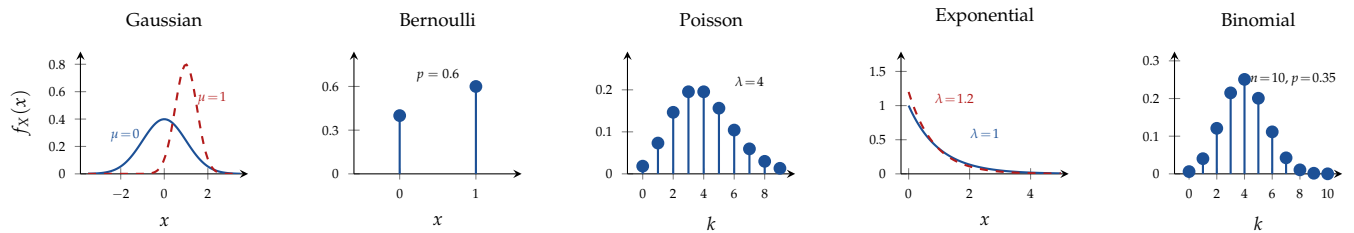


Figure 1: The five distributions introduced in this section. *Blue*: primary parameter choice; *dashed red*: alternative parameter.

## Classic distributions

THE FIVE DISTRIBUTIONS shown in Figure ?? appear throughout the course.

**Gaussian.**  $X \sim \mathcal{N}(\mu, \sigma^2)$  has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \quad (3)$$

It maximises entropy subject to fixed mean and variance, and is the attractor for sums of finite-variance variables (CLT).

**Bernoulli.**  $X \sim \text{Bernoulli}(p)$ :  $P(X = x) = p^x(1-p)^{1-x}$ ,  $x \in \{0, 1\}$ . Mean  $p$ , variance  $p(1-p)$ .

**Binomial.**  $X \sim \text{Binomial}(n, p)$ :  $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ . Mean  $np$ , variance  $np(1-p)$ .

**Poisson.**  $X \sim \text{Poisson}(\lambda)$ :  $P(X = k) = e^{-\lambda} \lambda^k / k!$ . Mean = variance =  $\lambda$ .

**Exponential.**  $X \sim \text{Exp}(\lambda)$ :  $f_X(x) = \lambda e^{-\lambda x}$ ,  $x \geq 0$ . The only continuous memoryless distribution.

### Generating functions

THREE GENERATING FUNCTIONS encode all statistical information in a distribution. They are the main calculational tools we will use to manipulate and identify distributions.

#### Moment generating function

The **moment generating function** (MGF) is

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx. \quad (4)$$

Expanding  $e^{tX} = \sum_{n=0}^{\infty} (tX)^n / n!$  and taking expectations term by term,

$$M_X(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n], \quad (5)$$

so the  $n$ -th moment is recovered by differentiation:

$$\mathbb{E}[X^n] = \left. \frac{d^n M_X}{dt^n} \right|_{t=0}. \quad (6)$$

For independent  $X, Y$ , the MGF of the sum factorises:  $M_{X+Y}(t) = M_X(t)M_Y(t)$ , since  $\mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$  under independence.

#### MGF of the Gaussian

For  $X \sim \mathcal{N}(\mu, \sigma^2)$ :

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx. \quad (7)$$

Combine the exponentials:

$$tx - \frac{(x - \mu)^2}{2\sigma^2} = -\frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}. \quad (8)$$

The constant terms factor out; the remaining integral is a Gaussian normalisation equal to 1:

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right). \quad (9)$$

Differentiating at  $t = 0$  gives  $\mathbb{E}[X] = \mu$  and  $\text{Var}(X) = \sigma^2$ .

#### MGF of the Bernoulli

$M_X(t) = (1 - p) + pe^t$ . Then  $M'_X(0) = p$  and  $\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = p(1 - p)$ .

#### Cumulant generating function

The **cumulant generating function** (CGF) is

$$K_X(t) = \log M_X(t), \quad (10)$$

and the  $n$ -th **cumulant** is  $\kappa_n = K_X^{(n)}(0)$ . The first two are  $\kappa_1 = \mathbb{E}[X]$  and  $\kappa_2 = \text{Var}(X)$ ; higher cumulants measure non-Gaussianity ( $\kappa_3$  is skewness,  $\kappa_4$  is excess kurtosis). All cumulants of a Gaussian beyond the second vanish.

WHY THE LOGARITHM? Because for independent  $X, Y$ ,

$$K_{X+Y}(t) = K_X(t) + K_Y(t). \quad (11)$$

Cumulants are additive for independent variables; moments are not. This makes the CGF the natural tool for large-system analysis.

#### CGF of the Gaussian

$K_X(t) = \mu t + \frac{1}{2}\sigma^2 t^2$ . The CGF is a polynomial of degree 2 — the hallmark of the Gaussian. Hence  $\kappa_1 = \mu$ ,  $\kappa_2 = \sigma^2$ ,  $\kappa_n = 0$  for  $n \geq 3$ .

#### CGF of the Poisson

The MGF is  $M_X(t) = e^{-\lambda} \sum_k (\lambda e^t)^k / k! = \exp(\lambda(e^t - 1))$ , so

$$K_X(t) = \lambda(e^t - 1), \quad \kappa_n = \lambda \quad \forall n \geq 1. \quad (12)$$

All cumulants equal  $\lambda$ , confirming  $\mathbb{E}[X] = \text{Var}(X) = \lambda$ .

#### Characteristic function

The **characteristic function** (Fourier transform of  $f_X$ ) is

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx, \quad t \in \mathbb{R}. \quad (13)$$

It always exists since  $|e^{itx}| = 1$ , and it *uniquely* determines the distribution even when the MGF does not exist (heavy tails). The inversion formula reconstructs the pdf:

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt. \quad (14)$$

**Gaussian.**  $\phi_X(t) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2)$ , obtained by substituting  $t \rightarrow it$  in (??).

Figure ?? summarises the relationships among the three generating functions.

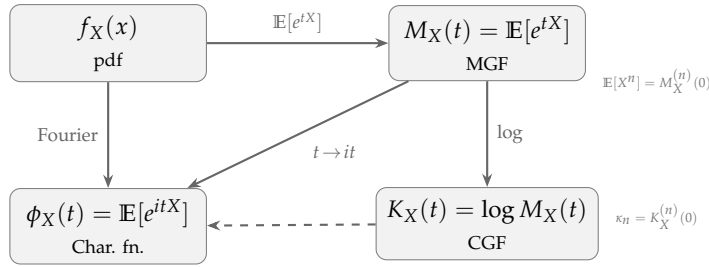


Figure 2: Relationships among the three generating functions. Arrows show how each is constructed from the previous.

### Multivariate distributions

FOR TWO RANDOM VARIABLES  $X, Y$  the **joint pdf**  $f_{X,Y}(x, y)$  satisfies  $\iint f_{X,Y} dx dy = 1$ . The **marginal** pdfs are

$$f_X(x) = \int f_{X,Y}(x, y) dy, \quad f_Y(y) = \int f_{X,Y}(x, y) dx. \quad (15)$$

The **conditional** pdf is  $f_{X|Y}(x|y) = f_{X,Y}(x, y) / f_Y(y)$ , which is Bayes' theorem in density form:  $f_{X|Y}(x|y) \propto f_{Y|X}(y|x) f_X(x)$ .

$X$  and  $Y$  are **independent** iff  $f_{X,Y} = f_X f_Y$ . The **covariance** and **correlation** are

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \quad \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1]. \quad (16)$$

### The multivariate Gaussian

A random vector  $\mathbf{x} \in \mathbb{R}^N$  follows  $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$  if

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^N \det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (17)$$

We derive this form from first principles in Section ??.

## Gaussian integrals

GAUSSIAN INTEGRALS are the fundamental computational engine of statistical physics and machine learning theory. The ability to evaluate them in closed form — even in hundreds of dimensions — is what makes Gaussian models analytically tractable.

### The one-dimensional integral

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2} dx = \sqrt{\frac{2\pi}{a}}, \quad a > 0. \quad (18)$$

**Derivation.** Let  $I = \int e^{-\frac{a}{2}x^2} dx$  and square it:

$$I^2 = \iint e^{-\frac{a}{2}(x^2+y^2)} dx dy. \quad (19)$$

Switch to polar coordinates  $x = r \cos \theta$ ,  $y = r \sin \theta$ , with area element  $r dr d\theta$ :

$$I^2 = 2\pi \int_0^{\infty} r e^{-\frac{a}{2}r^2} dr. \quad (20)$$

Substitute  $u = \frac{a}{2}r^2$  so  $r dr = du/a$ :

$$I^2 = \frac{2\pi}{a} \int_0^{\infty} e^{-u} du = \frac{2\pi}{a}, \quad (21)$$

giving  $I = \sqrt{2\pi/a}$ . The polar-coordinates trick is illustrated in the margin.

Adding a linear source term by completing the square,  $-\frac{a}{2}x^2 + bx = -\frac{a}{2}(x - b/a)^2 + b^2/(2a)$ , and shifting  $x \rightarrow x + b/a$ :

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}ax^2+bx} dx = \sqrt{\frac{2\pi}{a}} e^{\frac{b^2}{2a}}. \quad (22)$$

### The multidimensional integral

Let  $\mathbf{x} \in \mathbb{R}^N$ ,  $\mathbf{A}$  real symmetric positive-definite,  $\mathbf{b} \in \mathbb{R}^N$ . Then:

$$\int_{\mathbb{R}^N} d^N x \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}} \exp\left(\frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right). \quad (23)$$

**Step 1: Complete the square.** Claim:

$$-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} = -\frac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^T \mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}. \quad (24)$$

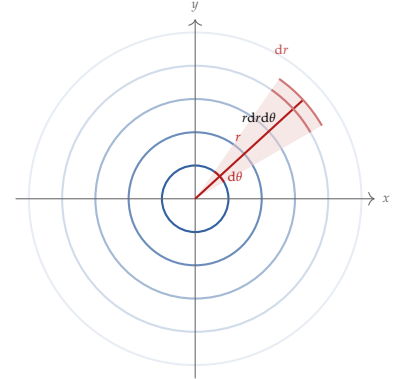


Figure 3: Squaring  $I$  converts the product of two 1-D Gaussians into a 2-D radially symmetric integral. The area element in polar coordinates is  $r dr d\theta$ .

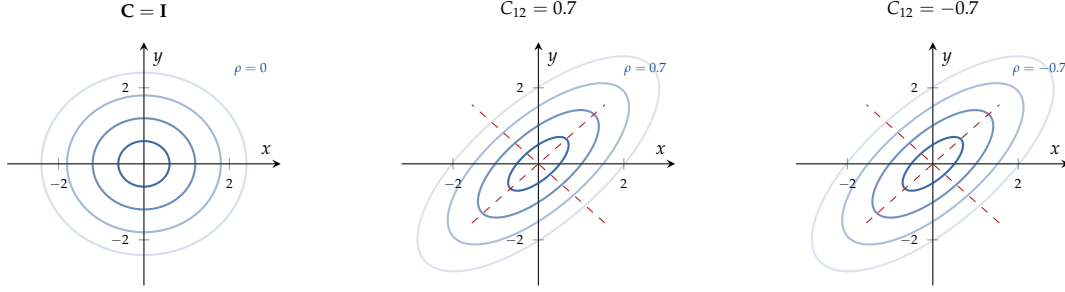


Figure 4: Level sets (constant-density contours) of 2-D Gaussians. Left: isotropic ( $C = I$ ), circular contours. Centre: positive correlation  $\rho = 0.7$ , ellipse tilted at 45. Right: negative correlation  $\rho = -0.7$ , ellipse tilted at -45. Dashed red lines mark the principal axes (eigenvectors of  $C$ ).

Expanding the right-hand side and using  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$  and symmetry of  $\mathbf{A}$ :

$$\begin{aligned} \text{RHS} &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \frac{1}{2}\mathbf{x}^T\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{x} \\ &\quad - \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} + \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b} \\ &= -\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}, \end{aligned} \quad (25)$$

confirming (??).

**Step 2: Shift the variable.** Set  $\mathbf{y} = \mathbf{x} - \mathbf{A}^{-1}\mathbf{b}$  ( $d^N y = d^N x$ ):

$$\int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}} \int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}}. \quad (26)$$

**Step 3: Diagonalise  $\mathbf{A}$ .** By the spectral theorem  $\mathbf{A} = \mathbf{O}^T\mathbf{\Lambda}\mathbf{O}$  with  $\mathbf{O}$  orthogonal and  $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$ ,  $\lambda_i > 0$ . Set  $\mathbf{u} = \mathbf{O}\mathbf{y}$ , so  $d^N u = d^N y$  and  $\mathbf{y}^T\mathbf{A}\mathbf{y} = \sum_i \lambda_i u_i^2$ . The integral factorises:

$$\int d^N y e^{-\frac{1}{2}\mathbf{y}^T\mathbf{A}\mathbf{y}} = \prod_{i=1}^N \sqrt{\frac{2\pi}{\lambda_i}} = \sqrt{\frac{(2\pi)^N}{\det \mathbf{A}}}. \quad (27)$$

Combining (??) and (??) gives (??).

*The source trick and moment generation*

Define the **generating functional**<sup>1</sup>

$$Z(\mathbf{b}) = \int d^N x e^{-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x}} = Z_0 e^{\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}}, \quad (28)$$

where  $Z_0 = Z(0) = \sqrt{(2\pi)^N / \det \mathbf{A}}$ . Since  $\partial e^{\mathbf{b}^T\mathbf{x}} / \partial b_i = x_i e^{\mathbf{b}^T\mathbf{x}}$ , differentiating and setting  $\mathbf{b} = \mathbf{0}$  extracts moments:

$$\langle x_i \rangle = \frac{1}{Z_0} \frac{\partial Z}{\partial b_i} \Big|_{\mathbf{b}=\mathbf{0}}. \quad (29)$$

**First moment.**  $\partial Z / \partial b_i = Z(\mathbf{b})(\mathbf{A}^{-1}\mathbf{b})_i$ . At  $\mathbf{b} = \mathbf{0}$ :  $\langle x_i \rangle = 0$ .

**Second moment.** Differentiating again:

$$\frac{\partial^2 Z}{\partial b_i \partial b_j} \Big|_{\mathbf{b}=\mathbf{0}} = Z_0 (\mathbf{A}^{-1})_{ij}, \quad (30)$$

<sup>1</sup> The name *generating functional* comes from field theory; here it is simply the Gaussian integral as a function of the source  $\mathbf{b}$ .

so

$$\langle x_i x_j \rangle = (\mathbf{A}^{-1})_{ij} = C_{ij}. \quad (31)$$

The two-point correlator equals the  $(i, j)$  entry of the inverse of the matrix in the exponent. We call  $C_{ij}$  the **propagator**. The level sets of 2-D Gaussians for different covariance structures are shown in Figure ??.

### Wick's theorem

WICK'S THEOREM gives an algorithmic rule for computing any moment of a Gaussian in terms of the propagator alone. It is the central combinatorial identity of perturbative field theory.

#### Statement

Let  $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ . Then:

- (1) All odd moments vanish.
- (2) All even moments satisfy:

$$\langle x_{i_1} x_{i_2} \cdots x_{i_{2n}} \rangle = \sum_{\text{pairings}} \prod_{\text{pairs } (j,k)} C_{i_j i_k}, \quad (32)$$

where the sum runs over all  $(2n - 1)!! = 1 \cdot 3 \cdot 5 \cdots (2n - 1)$  distinct ways to pair  $\{i_1, \dots, i_{2n}\}$  into  $n$  unordered pairs.

#### Derivation via the generating functional

With  $Z(\mathbf{b}) = Z_0 e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}}$ , expand:

$$e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} = \sum_{n=0}^{\infty} \frac{1}{n!} \left( \frac{1}{2} \sum_{i,j} b_i C_{ij} b_j \right)^n. \quad (33)$$

The  $m$ -th moment is  $\langle x_{i_1} \cdots x_{i_m} \rangle = \partial^m e^{\frac{1}{2} \mathbf{b}^T \mathbf{C} \mathbf{b}} / \partial b_{i_1} \cdots \partial b_{i_m} |_{\mathbf{b}=\mathbf{0}}$ .

For odd  $m = 2n + 1$ : every term in (??) has even powers of  $\mathbf{b}$ ; after  $2n + 1$  derivatives at least one  $b$  factor remains, which vanishes at  $\mathbf{0}$ . All odd moments vanish.

For even  $m = 2n$ : only the  $n$ -th term in (??) survives. Expanding it,

$$\frac{1}{n! 2^n} \sum_{i_1, j_1} \cdots \sum_{i_n, j_n} C_{i_1 j_1} \cdots C_{i_n j_n} b_{i_1} b_{j_1} \cdots b_{i_n} b_{j_n}. \quad (34)$$

Differentiating  $2n$  times matches each  $b_{k_l}$  to exactly one index; the  $\frac{1}{n! 2^n}$  prefactor precisely cancels the  $n! 2^n$  over-counting of pairings, leaving the sum over distinct pairings of products of  $C_{k_a k_b}$ . This is Eq. (??).

### Worked examples

#### Fourth moment

For  $X \sim \mathcal{N}(0, \sigma^2)$  there are  $(4 - 1)!! = 3$  pairings of  $\{1, 2, 3, 4\}$ :

- $(1, 2)(3, 4)$ : contributes  $\sigma^2 \cdot \sigma^2 = \sigma^4$ .
- $(1, 3)(2, 4)$ : contributes  $\sigma^4$ .
- $(1, 4)(2, 3)$ : contributes  $\sigma^4$ .

So  $\langle X^4 \rangle = 3\sigma^4$ . (The three pairings are illustrated in the margin.)

Verification via the MGF: differentiating  $e^{\frac{1}{2}\sigma^2 t^2}$  four times at  $t = 0$  gives  $3\sigma^4$ . ✓

#### Two-variable moment

For  $(x_1, x_2) \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ , the three pairings of  $(1, 1, 2, 2)$  give

$$\langle x_1^2 x_2^2 \rangle = C_{11}C_{22} + 2C_{12}^2. \quad (35)$$

For independent variables ( $C_{12} = 0$ ) this reduces to  $\langle x_1^2 \rangle \langle x_2^2 \rangle$ . ✓

#### Sixth moment

$(6 - 1)!! = 15$  pairings, each contributing  $\sigma^6$ , so  $\langle X^6 \rangle = 15\sigma^6$ .

### Saddle-point approximation

MANY INTEGRALS in statistical physics and machine learning take the form

$$I(N) = \int_{-\infty}^{\infty} dx e^{-Nf(x)}, \quad (36)$$

where  $N$  is a large parameter (number of neurons, spins, data points) and  $f(x)$  is smooth. As  $N \rightarrow \infty$ , the integrand concentrates exponentially near the minimum of  $f$ .

#### One-dimensional case

Let  $x^*$  minimise  $f$ :  $f'(x^*) = 0$ ,  $f''(x^*) > 0$ . Taylor expand:

$$f(x) = f(x^*) + \underbrace{f'(x^*)}_{0}(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2 + \dots \quad (37)$$

Substituting and evaluating the Gaussian integral:

$$I(N) \approx \sqrt{\frac{2\pi}{Nf''(x^*)}} e^{-Nf(x^*)}, \quad N \rightarrow \infty. \quad (38)$$

The cubic correction is  $\mathcal{O}(N^{-1/2}) \rightarrow 0$  since fluctuations are of order  $(x - x^*) \sim N^{-1/2}$ .

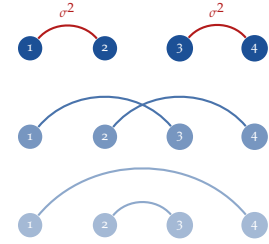


Figure 5: The three Wick pairings of  $\{1, 2, 3, 4\}$  for computing  $\langle X^4 \rangle$ . Each arc contributes a propagator  $\sigma^2$ .

Saddle-point concentration

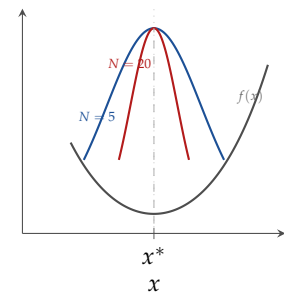


Figure 6: As  $N$  grows,  $e^{-Nf(x)}$  concentrates ever more sharply near  $x^*$ . The saddle-point approximation replaces  $f$  with its quadratic expansion around  $x^*$ .



### Multidimensional case

For  $\mathbf{x}^*$  with  $\nabla f(\mathbf{x}^*) = \mathbf{0}$  and Hessian  $\mathbf{H}_{ij} = \partial^2 f / \partial x_i \partial x_j|_{\mathbf{x}^*}$ :

$$\int_{\mathbb{R}^N} d^N x e^{-Nf(\mathbf{x})} \approx \left( \frac{2\pi}{N} \right)^{N/2} \frac{e^{-Nf(\mathbf{x}^*)}}{\sqrt{\det \mathbf{H}(\mathbf{x}^*)}}. \quad (39)$$

### Example: Stirling's approximation

From  $n! = \int_0^\infty x^n e^{-x} dx$ , substitute  $x = nu$ :

$$n! = n^{n+1} \int_0^\infty e^{-n(u - \log u)} du. \quad (40)$$

Identify  $f(u) = u - \log u$ . Saddle:  $f'(1) = 0$ ,  $f''(1) = 1$ ,  $f(1) = 1$ . Applying (??):

$$\log n! \approx n \log n - n + \frac{1}{2} \log(2\pi n). \quad (41)$$

### Example: Entropy of a spin system

The number of configurations of  $N$  spins with magnetisation  $m = \frac{1}{N} \sum_i \sigma_i$  is  $\Omega(m) = \binom{N}{N(1+m)/2}$ . Using Stirling with  $n_\pm = N(1 \pm m)/2$ :

$$s(m) \equiv \frac{\log \Omega}{N} = -\frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2}. \quad (42)$$

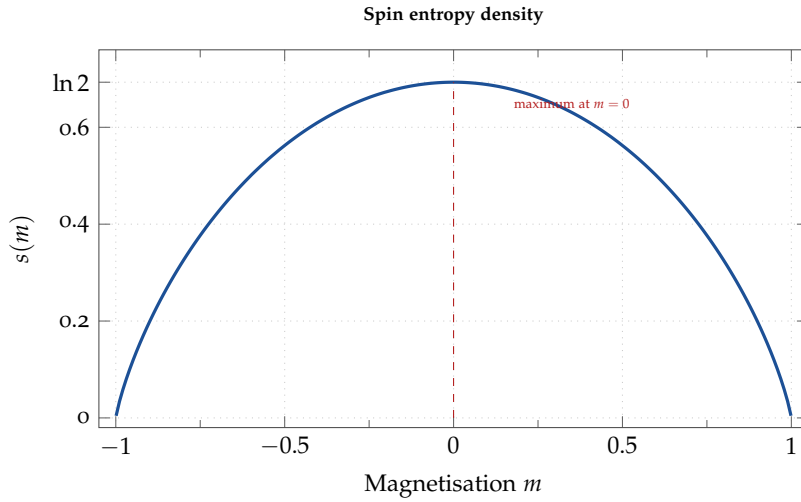


Figure 7: The spin entropy density  $s(m)$  from Eq. (42). It is maximised at  $m = 0$  (maximum disorder,  $s = \log 2$ ) and vanishes at  $m = \pm 1$  (all spins aligned).

### Hubbard–Stratonovich transformation

THE HUBBARD–STRATONOVICH (HS) TRANSFORMATION is the central algebraic trick of mean-field theory. It converts an intractable quadratic coupling between many degrees of freedom into an integral over a single auxiliary field, making the saddle-point approximation applicable.

*The basic identity*

$$e^{\frac{a^2}{2}} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{z^2}{2} + az}. \quad (43)$$

**Proof.** Complete the square in  $z$ :  $-\frac{z^2}{2} + az = -\frac{1}{2}(z - a)^2 + \frac{a^2}{2}$ . Then:

$$\frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + az} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \int e^{-\frac{(z-a)^2}{2}} dz = \frac{e^{a^2/2}}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = e^{a^2/2}. \quad \checkmark \quad (44)$$

*General rescaled form*

For  $\alpha > 0$  and coupling  $J$ , substituting  $a = J/\sqrt{\alpha}$  and  $z \rightarrow \sqrt{\alpha}w$  in (43):

$$e^{\frac{J^2}{2\alpha}} = \sqrt{\frac{\alpha}{2\pi}} \int_{-\infty}^{\infty} dz e^{-\frac{\alpha}{2}z^2 + Jz}. \quad (45)$$

**Derivation.** Start from (43) with  $a = J/\sqrt{\alpha}$ . Substitute  $z = \sqrt{\alpha}w$ :

$$e^{\frac{J^2}{2\alpha}} = \frac{1}{\sqrt{2\pi}} \int e^{-\frac{z^2}{2} + \frac{Jz}{\sqrt{\alpha}}} dz = \frac{\sqrt{\alpha}}{\sqrt{2\pi}} \int e^{-\frac{\alpha w^2}{2} + Jw} dw. \quad (46)$$

*Decoupling pairwise interactions*

The key application: a Boltzmann weight  $e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2}$  couples every pair of spins. Setting  $\alpha = N\beta J$  and  $J_{\text{eff}} = \beta J \sum_i \sigma_i$  in (45):

$$e^{\frac{\beta J}{2N}(\sum_i \sigma_i)^2} = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp\left(-\frac{N\beta J}{2}m^2 + \beta Jm \sum_{i=1}^N \sigma_i\right). \quad (47)$$

After this transformation the exponent is *linear* in each  $\sigma_i$ : the spins see only the uniform external field  $\beta Jm$  and are completely decoupled from one another.

*Application: Curie–Weiss mean-field theory*

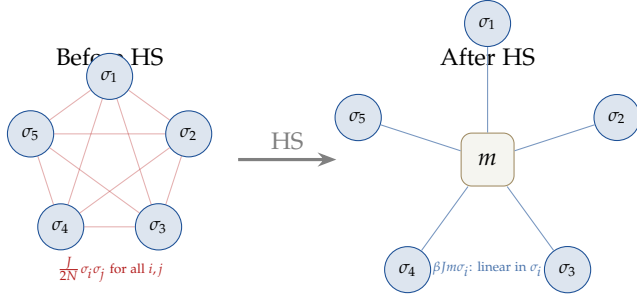


Figure 8: The Hubbard–Stratonovich transformation. Left: the original problem has all-to-all quadratic interactions  $J\sigma_i\sigma_j/2N$ , making the partition function a sum of  $2^N$  entangled terms. Right: after introducing the auxiliary field  $m$ , each spin interacts only with  $m$  through a *linear* coupling  $\beta Jm\sigma_i$ ; the spins decouple and the sum factorises.

THE CURIE–WEISS MODEL is the fully-connected Ising model:

$$H = -\frac{J}{2N} \sum_{i,j} \sigma_i \sigma_j = -\frac{J}{2N} \left( \sum_i \sigma_i \right)^2, \quad \sigma_i \in \{+1, -1\}. \quad (48)$$

The partition function is  $Z = \sum_{\{\sigma_i\}} e^{-\beta H}$ .

**Step 1 — Apply HS.**

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm e^{-\frac{N\beta J}{2} m^2} \sum_{\{\sigma_i\}} \prod_{i=1}^N e^{\beta J m \sigma_i}. \quad (49)$$

**Step 2 — Factorise.** Spins are independent after HS:

$$\sum_{\{\sigma_i\}} \prod_i e^{\beta J m \sigma_i} = \prod_i (e^{\beta J m} + e^{-\beta J m}) = (2 \cosh \beta J m)^N. \quad (50)$$

**Step 3 — Write as single integral.**

$$Z = \sqrt{\frac{N\beta J}{2\pi}} \int dm \exp(-Nf(m)), \quad f(m) = \frac{\beta J}{2} m^2 - \log(2 \cosh \beta J m). \quad (51)$$

**Step 4 — Saddle-point.**  $f'(m^*) = \beta J[m^* - \tanh(\beta J m^*)] = 0$ :

$$m^* = \tanh(\beta J m^*). \quad (52)$$

This is the **mean-field self-consistency equation**. The auxiliary field  $m$  is the magnetisation  $m^* = \langle \sigma_i \rangle$ .

**Step 5 — Phase structure.** Non-trivial solutions  $m^* \neq 0$  require the slope of  $\tanh(\beta J m)$  at  $m = 0$  to exceed 1:

$$\beta J > 1 \iff T < T_c \equiv J. \quad (53)$$

Above  $T_c$  only the paramagnetic solution  $m^* = 0$  exists. Below  $T_c$  two ferromagnetic solutions  $m^* = \pm m_0$  appear — this is the **ferromagnetic phase transition**, illustrated in Figure ?? . The entire derivation (HS  $\rightarrow$  spin factorisation  $\rightarrow$  saddle-point  $\rightarrow$  self-consistency equation) is the prototype for every mean-field calculation in this course, including the Hopfield model and the replica method.

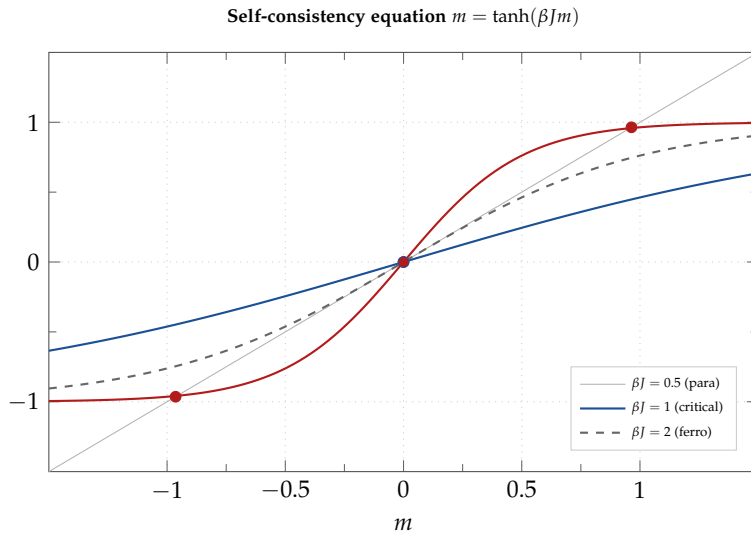


Figure 9: Graphical solution of the self-consistency equation (??). Intersections of  $y = \tanh(\beta J m)$  with the diagonal  $y = m$  give the solutions  $m^*$ . Below  $T_c$  ( $\beta J > 1$ , red curve) three intersections appear:  $m^* = 0$  (unstable) and  $m^* = \pm m_0$  (stable, filled dots). Above  $T_c$  ( $\beta J < 1$ , blue) only  $m^* = 0$  exists.